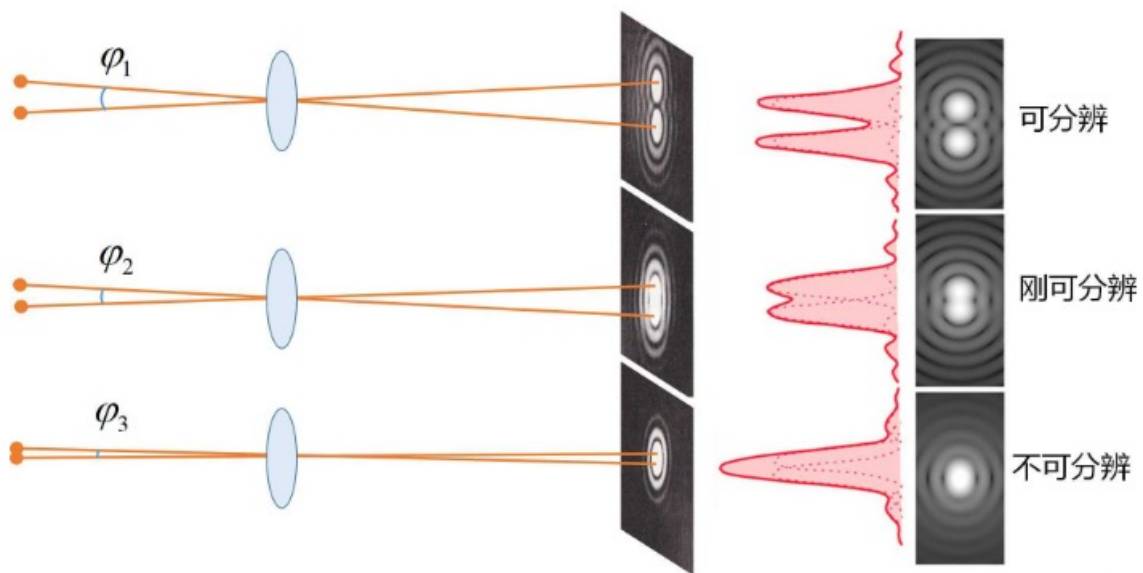
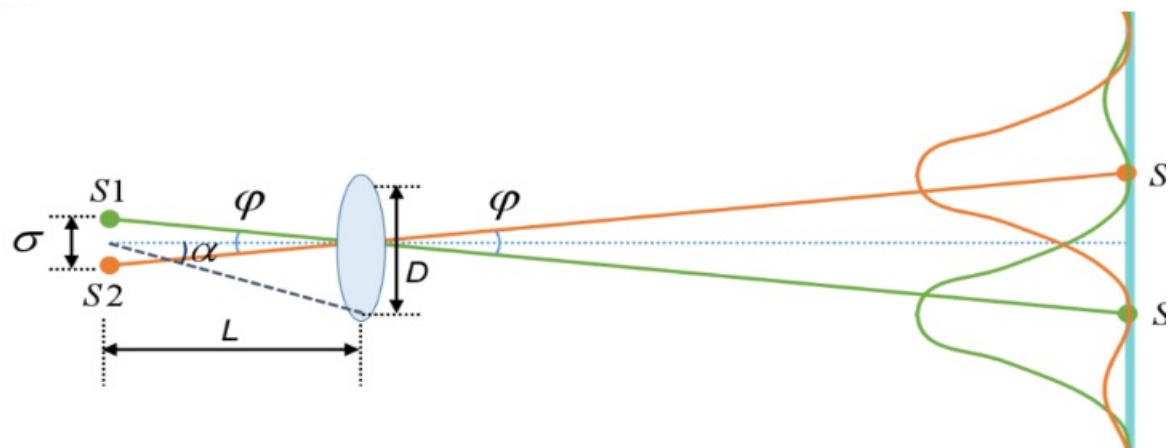
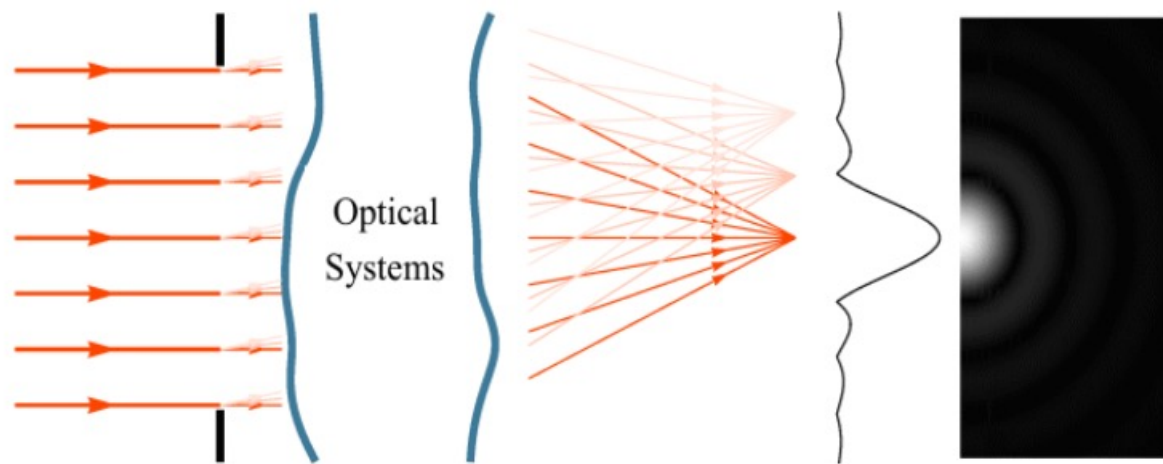


HomeWork

HW-1



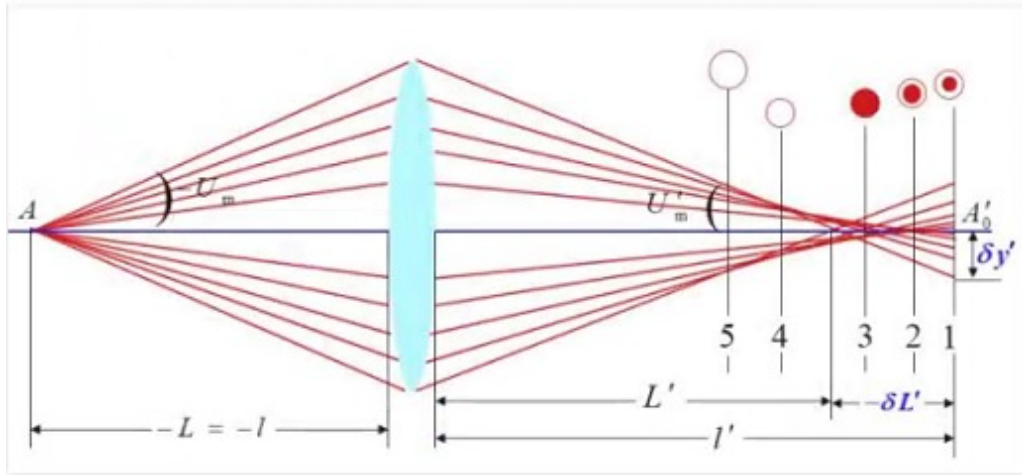
$$\varphi = \frac{1.22\lambda}{D} \quad \text{Angular resolution}$$

$$\sigma = \varphi L = \varphi f = \frac{1.22\lambda f}{D} \quad \frac{D}{2f} \approx \sin\alpha$$

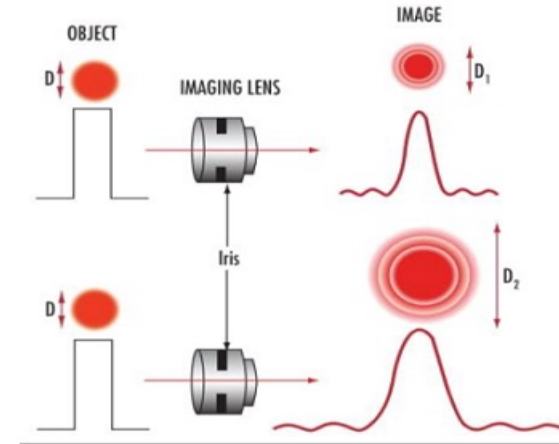
$$\sigma = \frac{1.22\lambda}{2n\sin\alpha} = \frac{0.61\lambda}{NA} \quad NA = n\sin\alpha,$$

HW-2

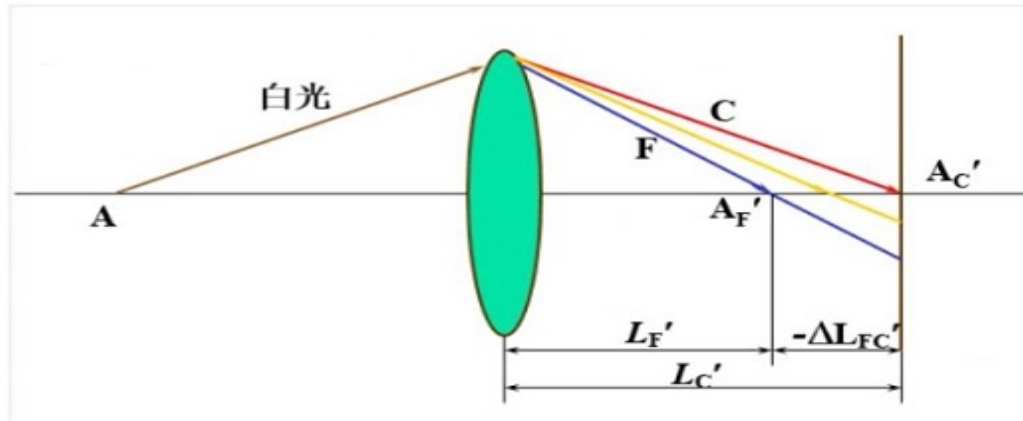
1. Spherical aberration



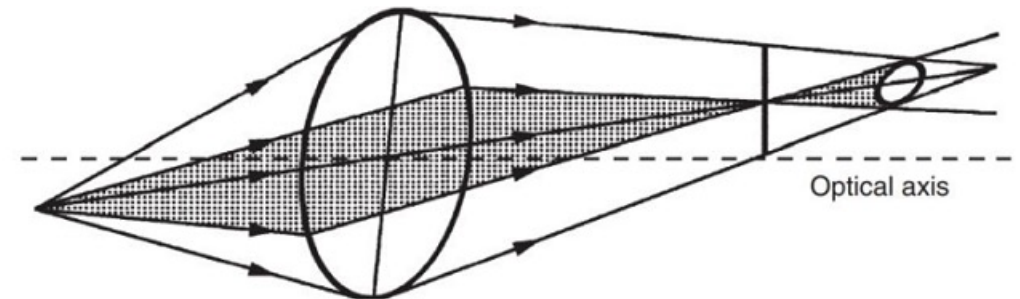
3. Aperture diffraction



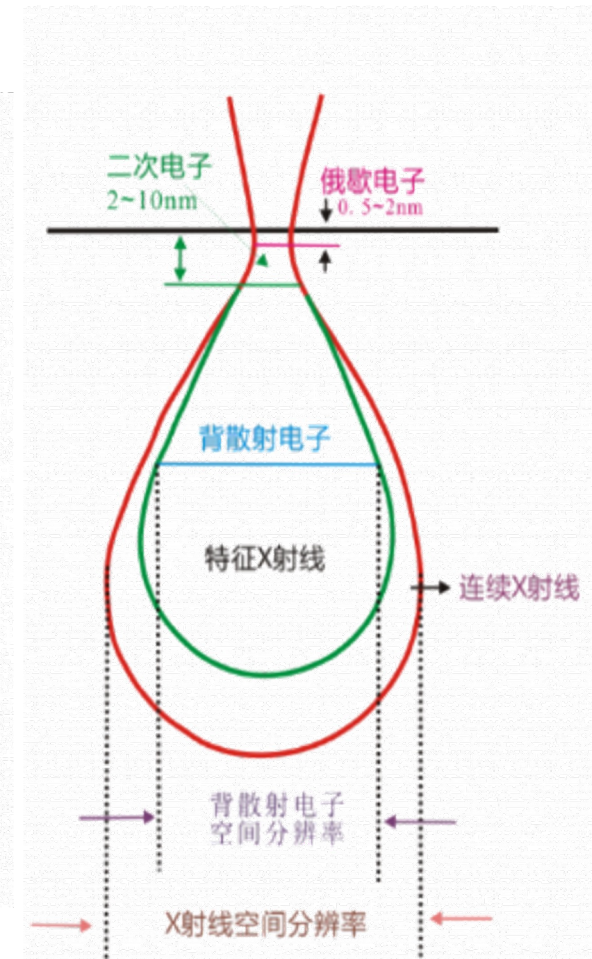
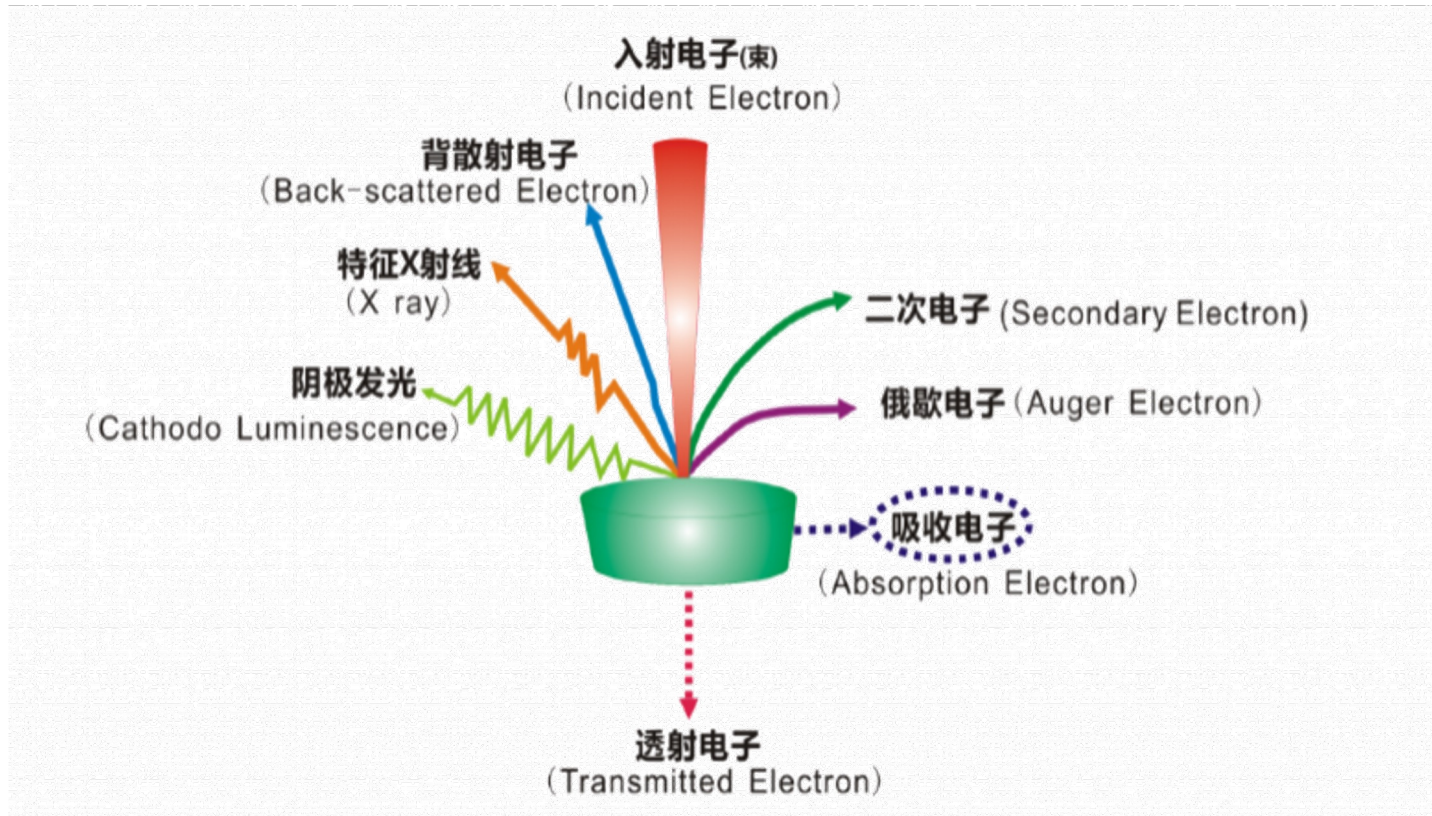
2. Chromatic aberration



4. Astigmatism



HW-3



Secondary electrons (50eV, 2-10nm)

Cathode fluorescence

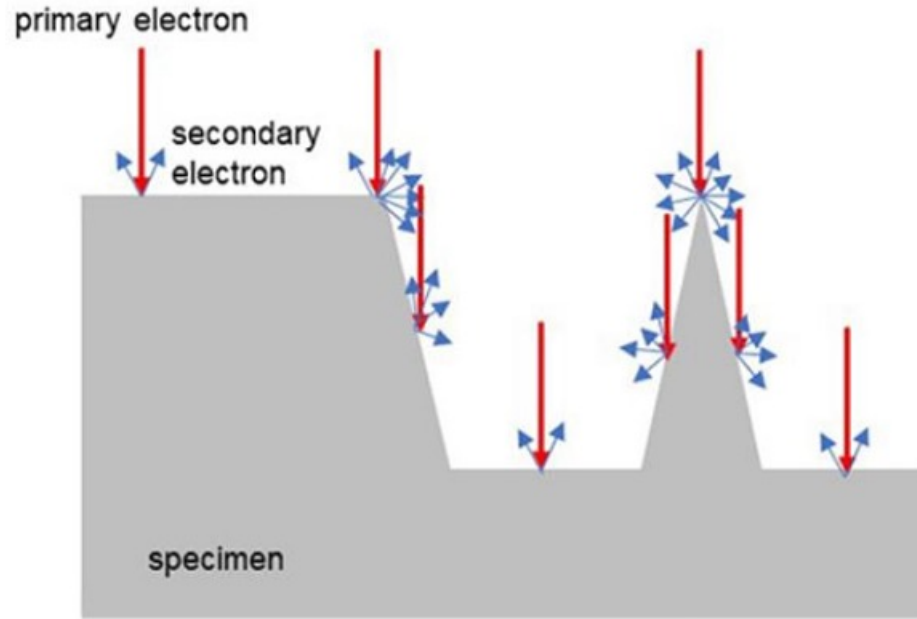
Backscattered electrons (50eV, 10^2 - 10^3 nm)

Characteristic X-rays (Characteristic energy, 10^3 nm)

Auger electrons (Characteristic energy, 3-5Angstrom)

Absorption electron

HW-3



SE can only be detected when they escape from specimen surface. At edges, SE are easier to escape. SE are emitted much more from the tip of a protrusion and the edge of a step than those from flat regions when the incident electrons scatter in the specimen. So more SE are collected at edges. It will make the image at edges extremely bright, which is called edge effect

	EELS	EDS
Composition	Uniform efficiency with Z 	Yield varies with Z – limited by absorption effects
Unknowns elements	Good at <u>verifying</u> unknowns	Good for <u>identifying</u> unknowns
Varying thickness	Works well down to $t = 0$ High background in very thick areas	Works well in thick areas of sample Low signal for thin areas
Chemical mapping	Chemical state directly revealed (ELNES)	No Chemical state information
Material Properties	Optical, Alloying, Charge transfer, Band gap, ...	Correlate with composition measurements
Spatial Distribution	No secondary excitation effects	Good for average composition

HW-5,6

5.计算高真空下气体分子的平均自由程

6.计算电子在某一速度下激发的X射线的最短波长